Marmara Üniversitesi
Mühendislik Fakültesi
Elektrik-Elektronik Mühendisliği

Laboratory Manual

EE 3011
Electronics I

Experiment (1)
Introduction to LTspice
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Experiment (1)
Introduction to LTspice

Objectives

- To simulate simple passive circuits using DC and Transient analysis.
- To be able to generate LTspice results using plots for various circuit variables.

Pre-lab Work

Before starting this experiment, you need to study the document "Introduction to LTspice IV with Examples", by Nicholas Lombardo, Linear Technology, 2015.

Circuits

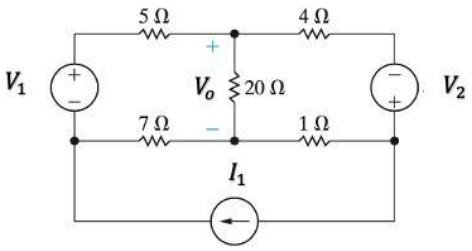


Figure 1: A DC circuit.

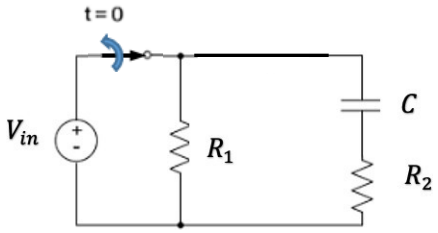


Figure 2: Transient circuit.

Procedure

Part (1): DC Analysis.

1. Connect the circuit in figure 1. Use the voltage and current sources values given by the lab instructor.
2. Using simulation, find the DC voltages and currents for all resistors. Record results in table 1.
3. Now make source V_1 as a variable voltage source and use the DC Sweep function to produce the transfer function curve V_o versus V_1.

Part (2): Transient Analysis.

1. Connect the circuit in figure 2. Use the circuit elements values given by the lab instructor. Set V_{in} as a periodic rectangular signal with frequency f and maximum amplitude V_{max}. The switch is initially closed.
2. Simulate the steady-state response of the circuit. Produce the steady-state voltage waveforms (versus time) across the capacitor C and the resistor $R_($. Put these curves together with the V_{in} curve on the same plot.
3. Now, as the switch is opened at t=0, simulate the circuit using the Transient analysis function and generate the voltage waveform (versus time) across the capacitor C and the resistor R_2.

Put these curves together with the V_{in} curve on the same plot. Also observe the waveformof the current through $R_($. From the curves, find the time constant of the circuit.

4. Change the value of f and repeat step 3.

Results and Analysis

Part (1): DC Analysis.

Table 1. DC Simulations in Part (1).

$V_1 = 10V$ $V_2 = 5V$ $I_1 = 3A$	Simulation	Calculation	Difference %
Voltage across 1 Ω resistor	0.81 V	0.81 V	%0
Current through 1 Ω resistor	0.81 A	0.81 A	%0
Voltage across 4 Ω resistor	8.75 V	8.57 V	%0.0018
Current through 4 Ω resistor	2.19 A	2.19 A	%0
Voltage across 5 Ω resistor	11.69 V	11.88	%0.0019
Current through 5 Ω resistor	2.33 A	2.34 A	%0.0001
Voltage across 7 Ω resistor	4.67 V	4.64 V	%0.0003
Current through 7 Ω resistor	0.6625 A	0.6625 A	%0
Voltage across 20 Ω resistor	3.02 V	2.7625 V	%0.003
Current through 20 Ω resistor	0.1475 A	0.1475 A	%0

1. Comment on the difference between simulation and theoretical values in table 1.

All voltage and current differences are less than 0.003%. The simulation results are very close to the results I calculated using the node voltage method. It can be said that they are the same. I think we can ignore this error in practical implementation.

Part (2): Transient Analysis.

Table 2. Transient Simulations in Part (2).

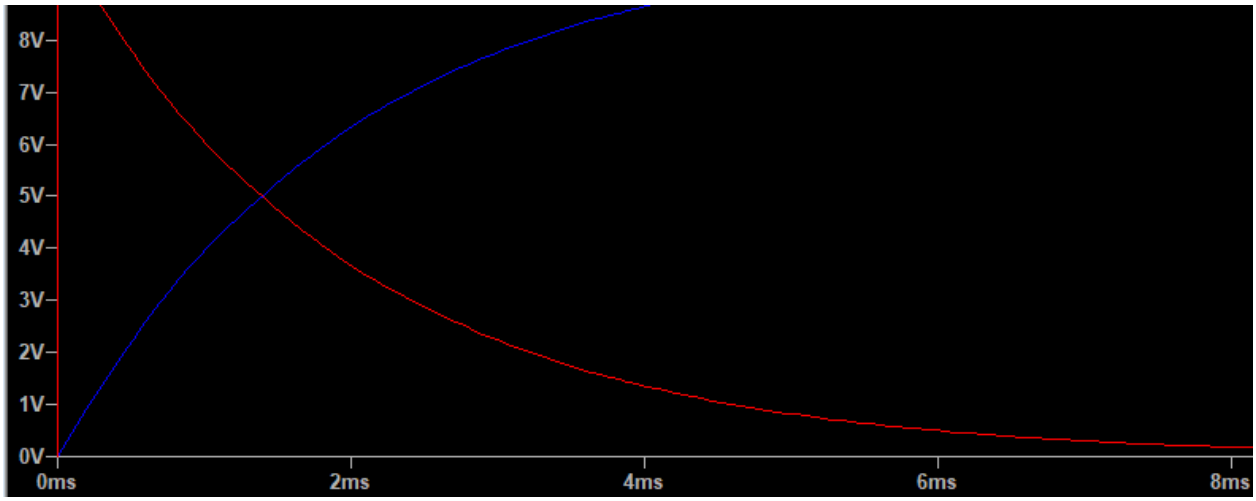
$R_1 = 1\text{k}\,\Omega$ $R_2 = 2\text{k}\,\Omega$ $C = 1\mu\text{F}$	Simulation	Calculation	Difference %
Circuit time constant	2 ms	0.66 ms	0.01

1. Comment on the difference between simulation and theoretical values in table 2.

   The difference is %0.01, which I think is negligible.

2. State your observations on the simulated behavior of the given RC circuit, including the effect of changing the input frequency.

   Before t_0=0, there is no change in voltage and current in the circuit because the capacitor is empty. After turning on the switch, the voltage curves of C (blue) and R2 (red) are as follows:



The capacitor charges itself, and the voltage across the resistor connected to the capacitor drops exponentially. The time constant is approximately 2 ms. If I change the frequency and adjust the T_{period} and T_{on} values, I get the same microsecond value. The formula is R*C and we expect the same result.